Modeling a Fine-Tuned Deep Convolutional Neural Network for Diagnosis of Kidney Diseases from CT Images

Sohaib Asif[1,2]
[1]School of Electronic Science and Engineering
Xi'an Jiaotong University, China
[2]School of Computer Science and Engineering
Central South University, China
sohaibasif@csu.edu.cn

Yi Wenhui*, Si Jinhai
School of Electronic Science and Engineering
Faculty of Electronic and Information Engineering
Xi'an Jiaotong University, China
*yiwenhui@mail.xjtu.edu.cn
jinhaisi@mail.xjtu.edu.cn

Qurrat ul Ain
School of Public Health
Central South University, China
annie_12310@yahoo.com

Yi Yueyang[1], Hou Jin[2]
[1]Health Science Center
Xi'an Jiaotong University, China
05115098@qq.com
[2]School of Basic Medical Science
Xi'an Medical University, China
1127538475@qq.com

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***Abstract*— Kidney abnormalities are common in the world and due to the scarcity of urologists, there is a need to develop an automated system for the detection of kidney diseases. In this paper, we present a powerful and unique deep transfer learning (TL) architecture based on a pre-trained VGG19 model and a naïve Inception module to efficiently detect major kidney diseases from CT images. We customized the architecture of the VGG19 model by removing the fully connected layers and placing a randomly initialized naïve Inception module and other layers such as batch normalization, dropout and dense layers to avoid vanishing gradient and overfitting problems and to improve performance. We consider two ways of TL: feature extractor and fine-tuning, which is technically implemented and evaluated. Grad-CAM and feature map techniques are used to visualize regions of interest that are meaningful for the diagnosis of kidney disease. 4000 kidney CT images were used to evaluate the performance of the proposed model using different performance metrics. Experimental results show that the proposed model exhibits superior performance after applying fine-tuning compared to other method. The results showed that the proposed model with a fine-tuning strategy produced an excellent accuracy of 99.25% in detecting three different kidney diseases from CT images.**

Keywords— Transfer learning, VGG19, Kidney abnormalities, Deep learning, Kidney stone detection.

I. INTRODUCTION

Kidney disease is a common public health problem that affects people of all ages and genders. The development of renal tumors, cysts and stones leads to chronic renal disease with adverse consequences such as renal dysfunction and premature death. Renal cysts are round pouches and closed pockets filled with fluid on or in the kidneys [1]. Kidney stones are hard objects made up of chemicals and salts in the kidneys that block the ureters [2]. A kidney tumor is the most common type of cancer that starts in the kidney and can be malignant or benign [3]. These kidney diseases do not cause significant symptoms in the early stages and lead to a gradual decline in kidney function over time. Early detection of kidney disease is critical to prevent the development of chronic disease and kidney failure. Large-scale routine screening of patients with kidney disease using different medical devices generates a large amount of medical data. Examination of patients with computed tomography (CT) scan is ideal as it offers high contrast images and 3D information [4]. Manual examination of medical images is costly and time consuming due to the lack of nephrologists and medical specialist worldwide, which can lead to misdiagnosis. Medical image analysis plays an important role in the diagnosis of different diseases, but the need for a second opinion due to the limited number of health personnel greatly affects the process. Therefore, accurate diagnosis of patients with kidney disease is an absolute necessity.

Convolutional Neural Networks (CNNs) are a class of deep learning (DL) technique that are one of the effective methods in the medical field, such as brain tumor classification, skin cancer detection [27, 28], etc. CNNs represent a major breakthrough in medical image processing tasks, capable of automatically extracting richer and more robust features than traditional machine learning (ML) techniques. The initial layers of CNN are useful for extracting general features such as colors and blobs, while higher layers are useful for extracting domain-specific features [5]. However, full training of CNNs is unreliable, data-intensive, time-consuming, and prone to overfitting. The scarcity of labeled and annotated data in the medical field hinders the full utilization of CNNs. Full training of CNNs requires high computational resources and careful hyper-parameter optimization. Thus, training networks from scratch is prone to overfitting, tedious, data hungry, and requires a great deal of expertise [6]. Recently, many researchers have been working on the detection of kidney diseases, but due to data limitations, most of the previous methods used ML techniques [7] and some used ultrasound images [8]. The transfer learning (TL) strategy has become predominant to overcome problems such as data limitation, full network training. The TL strategy used pre-trained models with ImageNet weights, which can provide better performance with less data and reduced computation time compared to training the model from scratch. TL techniques are the most common practice and have been successful in many biomedical applications where medical data is scarce [9]. In view of the above, we propose a deep TL architecture for diagnosing kidney diseases from CT images. We have designed a highly efficient and end-to-end solution based on the already proven VGG19 architecture. The major research contributions of this study are as follows:

1. This study aims to design a novel deep TL architecture based on a pre-trained VGG19 model and a naïve Inception module to extract robust features and reduce computational complexity while improving the efficiency of the model. TL was employed in two different ways: as a feature extractor and as a fine-tuner to design an efficient model for kidney disease detection.
2. Batch normalization, dropout, flattening and augmentation are incorporated into the proposed model to reduce overfitting and yield remarkable results in detecting major kidney diseases.
3. Grad-CAM and feature maps techniques were used for the interpretability of the model. Through a rigorous evaluation, the proposed model showed promising results for the detection of kidney diseases.

II. Literature Review

Recently, numerous ML and DL methods have been proposed to detect renal disease. Aksakalli et al. [10] used six different ML algorithms and also used a CNN to detect kidney stones from X-ray images. They reported an f1 score of 85.30%. The authors [11] proposed a CNN model to detect ureteral stones. They also used a majority voting technique with 384 calcifications to train the model. They achieved an accuracy of 93%, which is higher than that of radiologists. The authors [12] proposed an ensemble model using three pre-trained architectures for feature extraction and SVM for classification of kidney stones using ultrasound images. Their method achieved 95.58% accuracy. The authors [13] used a pre-trained AlexNet model to extract features, which were then fed into an SVM to classify images as stone, cyst, and normal. They achieved 91.80% accuracy. Cui et al. [14] developed a DL and threshold-based model to detect kidney stones from CT images. They used 625 CT images to train the model and achieved 90.30% accuracy on the test set. The authors [15] propose a multi-feature fusion CNN to detect and localize kidney stones from CT images. They experimented with 3722 images and achieved 94.67%. Martin et al. [16] uses the DL method to detect ureteral stones from thin CT images. They used 465 thin-slice CT images and achieved 100% sensitivity. Parakh et al. [17] proposed a cascaded CNN to diagnose kidney stones from CT images. They used the TL strategy and used 535 CT images to train the model. Their method reported 95% accuracy. The authors [18] propose a 7-layer CNN architecture to detect kidney lesions from CT images. They collected 614 images from the clinic and conducted the experiment. The model reported 90.36% accuracy. Islam et al. [19] constructed a new kidney CT dataset and made it publicly available to researchers. They use three transformer variants and three pre-trained models for performance comparisons. They achieved 99.30% detection of kidney disease using the Swin Transformer. Fitri et al. [20] proposed a CNN model and used a total of 2430 CT images to classify stones, reporting 99.59% accuracy. Pavithra et al. [21] designed a CNN model and used the GLCM method for feature extraction. They also used fuzzy C-means clustering and achieved the highest accuracy of 98.30%.Yildirim et al. [22] proposed an end-to-end deep learning model to detect kidney stones from coronal CT images using the XResNet-50 architecture. They achieved 96.82% accuracy in classifying CT images as stone and healthy. Nithya et al. [7] uses artificial neural network and k-means clustering algorithm to classify kidney stones. They performed classification and segmentation and achieved 93.45% accuracy. Blau et al. [23] proposed a DL framework to detect renal cysts from CT images. The proposed framework achieves a true positive rate of 84.30%.

III. Materials and Methods

A. Data acquisition and preprocessing

In this work, kidney CT images were obtained from a publicly available repository called CT KIDNEY DATASET [19]. CT images collected from patients diagnosed with kidney disease including stones, tumors, normal, and cysts from different hospitals in Bangladesh. Due to limited computing power, 1000 CT images were selected from each class. In other words, 1000 images of kidney stones, 1000 images of cysts, 1000 images of tumors and 1000 images of normal, for a total of 4000 images. After collecting the dataset, the next step is data pre-processing which includes different methods such as resizing, splitting and augmentation. We first resized all the kidney CT images to 224×224 according to the input size requirements of the architecture. After that, the curated CT dataset was divided into training and testing, where 80% or 3200 CT images were used for training and 20% or 800 CT images were used for testing. Finally, this study uses data augmentation to synthetically increase the training samples to reduce overfitting problems and improve classification performance. We used three different augmentation strategies such as random contrast, rotation and flipping. Figure 1 shows the random CT samples collected from the curated dataset.

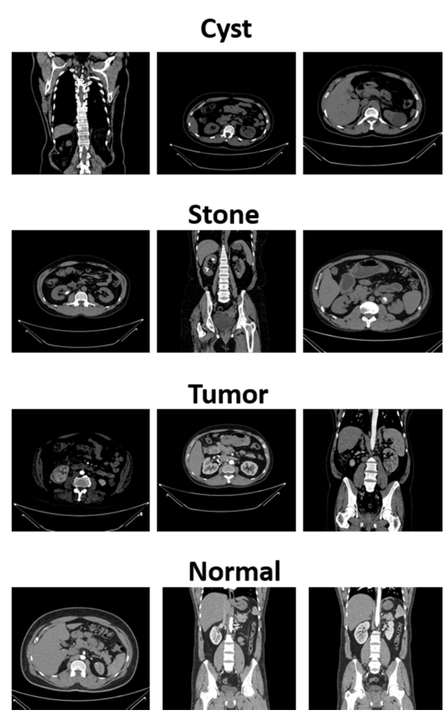

Figure 1. Random CT samples

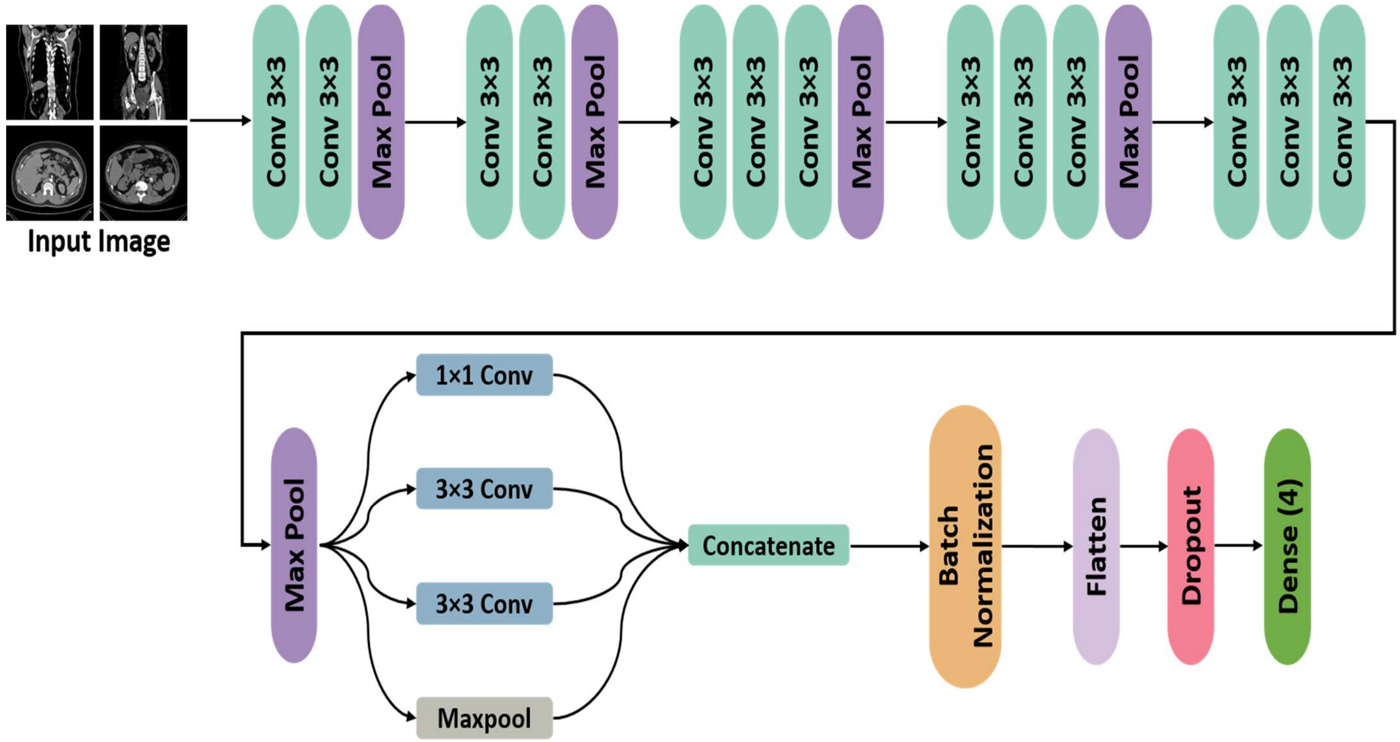

Figure 3. Main architecture of the proposed deep TL architecture for kidney disease detection

B. The proposed deep transfer learning architecture

The term depth in a deep CNN describes the stacking of layers in the model, which can learn low to high level features and can improve performance. Convolutional layers in CNN are responsible for extracting discriminative features, pooling layers reduce feature map size and computational complexity, and dense layers are responsible for classifying the extracted features. Instead of training a network from scratch, an alternative procedure is to use the TL strategy on already proven pre-trained models. In this research, we considered the VGG19 pre-trained model [24], a successful variant of VGGNet, to create a more robust and accurate deep TL model that effectively solves the kidney diseases classification problem. VGG19 consists of five convolution blocks with 19 layers stacked in succession. The first and second convolution blocks contain two 3×3 convolution layers, while the last three blocks contain two convolution layers each, with each block followed by a pooling layer. After the fifth block, there are three fully connected layers where the first two layers have 4096 neurons and the last layer has 1000 neurons to classify the images of the ImageNet dataset. However, VGGNET has various deployment problems, such as the last fully connected layer in VGG19, which increases the depth and number of parameters and is prone to the problem of vanishing gradients. In this work, we proposed a new deep TL architecture by modifying the VGG19 architecture to address the challenges seen in VGGNET. First, we remove the fully connected layers from the VGG19 architecture because these layers only increase the computational cost and parameters. After that, we added a naïve Inception module [25] to avoid the vanishing and exploding gradient problem. The Inception module was first introduced by GoogleNet [25] to reduce the overfitting and computational complexity problems while improving model efficiency. In addition, the 1×1 convolution module also makes the model computationally less expensive. We extracted robust and domain-specific features from the VGG19 model up to the block5 pooling layer and concatenated them with the naïve Inception module as shown in figure 2. The main reason for adding the naïve Inception module to the proposed deep TL architecture is to extract multi-level features, thereby reducing model complexity without affecting the feature extraction capability of the network. Furthermore, we added different random initialization layers such as flatten, batch normalization, dropout and dense layers after the naïve Inception module.

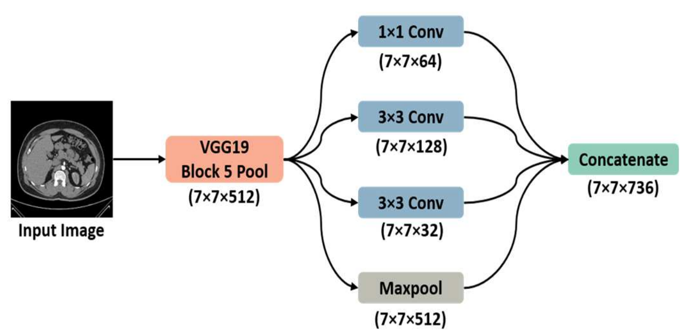

Figure 2. Architecture of the proposed deep TL model depicting the concatenation of a naïve Inception module with block5 max pooling layer of VGG19.

We have developed a new deep TL architecture based on a modification of the VGG19 architecture as shown in Figure 3. In addition to this, the features extracted from the block5 pool layers from VGG19 is concatenated with a naïve Inception module. The main reason for considering the naïve Inception module is to learn rich and complex features and make the model less computationally expensive. The naïve Inception module contains three different convolution layers with three different sizes of filters (1x1, 3x3, 5x5) and a ReLU activation function. Pooling is also performed. After that, batch normalization is added to ensure fast and accurate deep network training [26]. Further flatten layer is added to flatten the extracted features into 1-D of size 36064. A dropout factor of 0.4 is used to reduce overfitting. Finally, a dense layer of 4 softmax neurons is used to classify the CT images to get the final prediction.

We investigate two different TL strategies (feature extractor and fine-tuning) for the VGG19 model. First, we freeze the convolution base of the VGG19 architecture and initialize the model with pre-trained ImageNet weights. After that, the randomly initialized naïve Inception module and other layers are trained on the kidney dataset. To reduce the risk of destroying the feature extraction capability of the convolution base, we allow the gradient to back propagate within the new randomly initialized layers. Second, we employ a fine-tuning strategy on the deep TL architecture. After training the randomly initialized naïve Inception module and other layers, we unfreeze the 5th convolutional block of the VGG19 architecture and allow all layers in the 5th convolutional block to be fine-tuned with a smaller learning rate.

IV. RESULTS AND DISCUSSION

In this section, the hyper-parameters and implementation details of the proposed model are discussed. After that, experimental results using two different TL strategies, i.e., feature extractor and fine-tuner, are presented to investigate the efficacy of the proposed deep TL model using different performance metrics.

A. Hyper-parameters and implementation details

Our proposed deep TL architecture is trained using google Colab with Nvidia tesla T4 GPU and 12GB RAM. We use python with keras and tensorflow to import the model and instantiate it with the weights of ImageNet. All CT images in the dataset are resized to 224×224×3. After that, the curated CT dataset was split into training and testing, where 80% or 3200 CT images were used for training and 20% or 800 CT images were used for testing. In this study, we investigate two different TL strategies (feature extractor and fine-tuning) for the VGG19 model. For the feature extractor experiments, we froze the convolution base of the VGG19 architecture and initialized the model with pre-trained ImageNet weights. Afterwards, randomly initialized naïve Inception module and other layers are trained on the kidney dataset. To reduce the risk of destroying the feature extraction capability of the convolution base, we allow the gradients to be back-propagated within the new randomly initialized layers. For the fine-tuning experiments, after training a randomly initialized naïve Inception module and other layers, we unfrozen the 5th convolution block of the VGG19 architecture and allowed all layers in the 5th convolution block to be fine-tuned at a smaller learning rate. During training, the proposed model was trained for 50 epochs with an initial learning rate of 1e-2 for the feature extractor and a lower learning rate of 1e-3 for fine-tuning. The configured batch size is 32 and the model is trained using the Adam optimizer. The Adam optimizer allows for faster convergence and is very popular in medical imaging applications. Categorical cross-entropy was used as a loss function to train the model. The early stopping technique is used with patience = 15 epochs to stop training if validation loss does not improve. ReduceLROnPlateau is used with patience = 5, min_lr = 1e-6 and factor = 0.8.

B. Performance metrics

In this study, different performance evaluation measures are used to calculate the quantitative evaluation of the proposed deep TL architecture. False negative (FN), true positive (TP), true negative (TN), and false positive (FP) are used to calculate accuracy (1), recall (2), precision (3), and f1-score (4).

$$Accuracy = \frac{TN + TP}{TN + TP + FN + FP} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1 - Score = 2 \times \frac{Precision\ \times Recall}{Precision + Recall} \tag{4}$$

C. Result Analysis

In this study, to classify three major kidney diseases from CT images, we evaluated the performance of the proposed deep TL model as a feature extractor and fine-tuner using different performance metrics. A total of 3200 images were used for training and 800 images were used for testing. Table 1 shows the classification performance of the proposed model as a feature extractor and a fine-tuner. It can be seen that the proposed model provides a reasonable recall for cysts, normal, stone and tumors and provided the highest recalls of 1.0, 1.0, 0.99 and 0.98. From Table 1 we can see that the proposed model with fine-tuning scheme shows significant improvement and performs well in detecting kidney disease. Overall, the proposed architecture with the fine-tuning scheme exhibits significantly higher accuracy (99.25%) compared to the feature extractor experiment. In Figure 4, we analyze the confusion matrix of the proposed deep TL architecture using two different TL strategies (feature extractor and fine-tuning). The figure shows that 771 CT images are correctly classified with the feature extraction strategy and 794 CT images are correctly classified with the fine tuning strategy. The confusion matrix shows that the proposed architecture using the fine-tuning strategy has a low misclassification rate. This figure shows that all CT images of the cyst and norms were classified correctly. However, only two CT images of the stones and four CT images of the tumor were misclassified using the fine-tuning strategy. We can see that the proposed deep TL model with the feature extractor experiment

TABLE 1 CLASS-WISE PERFORMANCE OF THE PROPOSED MODEL USING FEATURE EXTRACTOR AND FINE TUNING EXPERIMENTS (ACC: ACCURACY, PRE: PRECISION, REC: RECALL, F1: F1-SCORE)

Experiment scenario	Overall Accuracy	Class	ACC (%)	PRE	REC	F1
Feature Extractor	96.37%	Cyst	97.87	0.92	0.99	0.96
		Normal	98.50	0.96	0.99	0.97
		Stone	97.00	0.99	0.89	0.94
		Tumor	99.37	0.98	0.99	0.99
Fine Tuning	**99.25%**	**Cyst**	**99.25**	**0.97**	**1.0**	**0.98**
		Normal	**100**	**1.0**	**1.0**	**1.0**
		Stone	**99.75**	**1.0**	**0.99**	**1.0**
		Tumor	**99.50**	**1.0**	**0.98**	**0.99**

achieves the lowest performance compared to the fine-tuning experiment. From the analysis, it can be seen that the proposed model shows a significant improvement in performance when we add the last convolutional block in the fine-tuning process. These findings strongly support the use of fine-tuning, which is essential for most medical imaging applications. The possible reason is that fine-tuning the last convolutional block of the proposed model retains domain-specific and relevant features that are closely related to our task. However, the proposed model with feature extractor experiments fully retains general features rather than domain specific features. The proposed model with a fine-tuning strategy shows a dramatic improvement in performance compared to the feature extractor experiments.

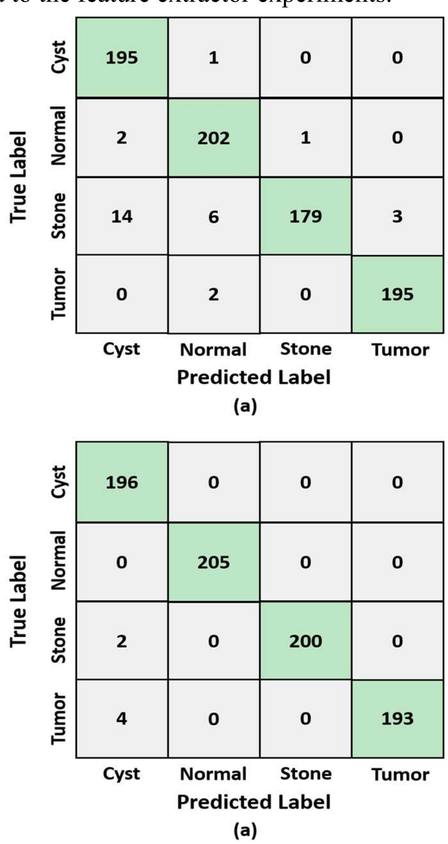

Figure 4. Confusion matrix of the proposed deep TL architecture (a) Feature extractor, (b) Fine tuning

D. Grad-CAM and feature map visualization

For better interpretability, in this section we explore feature maps and a Grad-CAM method for extracting gradients from the last convolutional layer of the proposed deep TL model to visualize regions of arbitrarily chosen input samples that are important for accurate predictions. Grad-CAM is a technique to visualize the critical areas in the images that the model uses for diagnosis. Figure 5 illustrates the Grad-CAM visualization of the proposed deep TL model using CT images from the test set. Red circles are used to highlight infected areas in CT images to compare the performance of the proposed deep TL model with expert opinion. It can be seen from the figure that the generated heatmaps focus more on the infected part of the CT images and draw the relevant classification results. Figure 6 shows the feature maps of the proposed deep TL architecture simply iterated from the input image to the output layer. As can be seen in Figure 6, the proposed deep TL model provides great insight and efficiently captures more specific features from CT images. Thus, our proposed model can solve the problem of detecting major kidney diseases in CT images.

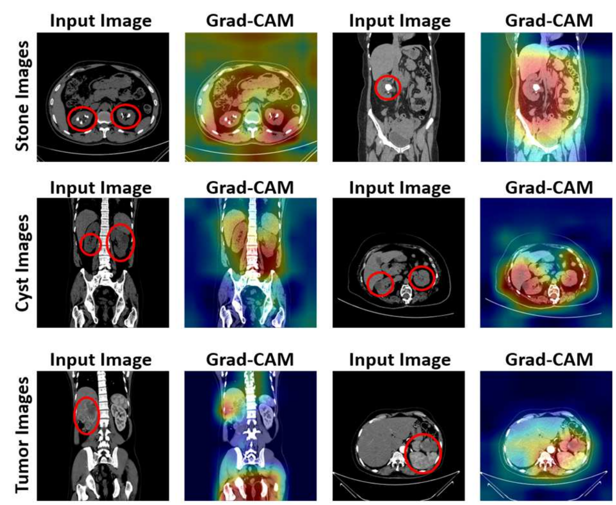

Figure 5. Grad-CAM analysis of the proposed deep TL model

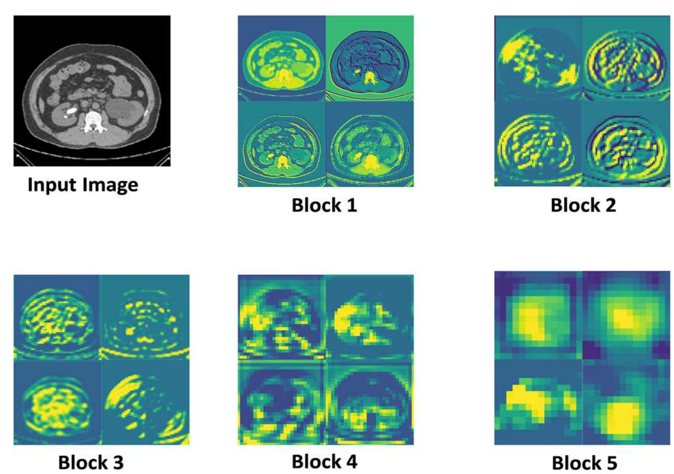

Figure 6. Feature map visualization of proposed model with different convolutional blocks

V. CONCLUSION

This study aims to design a novel and powerful deep TL architecture that is based on a pre-trained VGG19 model and a naïve Inception module to extract robust and discriminative features that reduce computational complexity while improving model efficiency. We customized the architecture of the VGG19 model by removing the fully connected layer and placing a naïve Inception module with random initialization and other layers such as batch normalization, dropout and dense layers to avoid vanishing gradient and overfitting problems and to improve robustness and generalization. TL was employed in two different ways: as a feature extractor and as a fine-tuner to design an efficient model for kidney disease detection. The Adam optimizer is used to demonstrate fast learning, and the cross-entropy loss function is used to minimize the loss in the training. In this work, Grad-CAM methods and feature maps were used for the interpretability of the proposed model. Our experimental results show that the proposed model with fine-tuning strategy achieves better classification performance and helps to improve overall robustness compared to feature extractor experiment. The results of this study suggest that the proposed model will be of great benefit to urologists in detecting kidney disease.

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